

UNIVERSITY OF CENTRAL PUNJAB



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Course Title: Robotics Design Lab - 1
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Project Report

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Program Objective: -	Course Objective: -	Course Learning Objective: -
TO BE FILLED IN BY THE STUDENT		
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Project Title:

"Line Follower Robot Using IR Sensors"
Delving deeper within the complexities of robotics.

“Project Report”

❖ Points to cover:

- Project Introduction
 - Components List
 - Block Diagram
 - Working Principle
 - Project Source Code
 - Code Explanation
 - Applications
 - Conclusion
-

❖ Project Introduction:

In the modern age of automation and robotics, autonomous mobile robots play a vital role in streamlining various industrial, commercial, and educational applications. One such foundational yet significant autonomous system is the “**Line Follower Robot**” a robot designed to detect and follow a predetermined path, usually represented by a black line on a white surface. These types of robots are commonly used in material handling, warehousing, and production lines where consistent and reliable movement along fixed paths is essential.

This project focuses on the design and implementation of a Line Follower Robot that utilizes a five-channel TCRT5000 infrared (IR) sensor array to detect and track the black line path. The robot’s movement is controlled by an Arduino Nano microcontroller, which reads sensor inputs and makes real-time decisions to control the direction and speed of two N20 geared DC motors through an L298N H-Bridge motor driver.

The IR sensors function by detecting the reflection of infrared light, black surfaces absorb more IR and are thus detected as low signals, whereas white surfaces reflect IR and are read as high signals. The robot interprets these signals to determine whether to move forward, turn slightly left or right, make sharp turns, or stop altogether when the line is lost.

This project not only enhances understanding of embedded systems and motor control but also demonstrates the practical integration of sensors, actuators, and decision-making algorithms. The simplicity of the design, combined with its practical significance, makes it a valuable educational tool and a stepping stone for more advanced robotics applications.

❖ Components List:

Following is the detailed components list for this line follower robot:

- **Arduino Nano:**

The microcontroller that serves as the brain of the robot. It processes input signals from the IR sensors and sends control commands to the motor driver to move the robot accordingly.



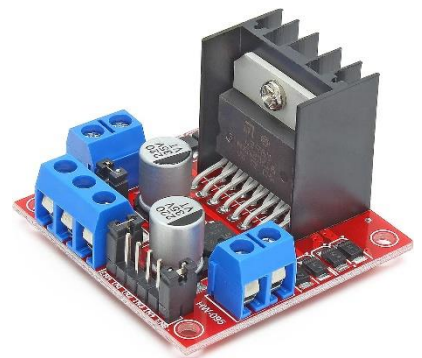
- **TCRT5000 IR Sensor Array (5-Channel):**

A module containing five infrared transmitter-receiver pairs used to detect the line beneath the robot. Each sensor provides a digital signal LOW when it detects a black surface (line) and HIGH for a white surface. These readings help the robot determine its direction.



- **L298N Motor Driver Module:**

A dual H-bridge motor driver IC used to control the speed and direction of two DC motors independently. It acts as an interface between the low-power control signals of the Arduino and the higher-power demands of the motors.



- **N20 DC Gear Motors (x2):**

Small-sized, high-torque gear motors used to drive the robot's wheels. Their geared-down output provides better control and slower, more powerful movement ideal for line-following applications.



- **Lithium Rechargeable Cells 3.7V(x2):**

Provides the necessary power to run the motors and the control circuit. It is connected through the motor driver to supply power both to the motors and the Arduino board.



- **Power Switch:**

A basic ON/OFF toggle switch integrated between the battery and the power circuit, used to control when the robot is powered without unplugging wires.



- **Wheels (x2):**

Attached to the motor shafts, the wheels convert motor rotation into forward or turning motion, enabling the robot to move along the path.



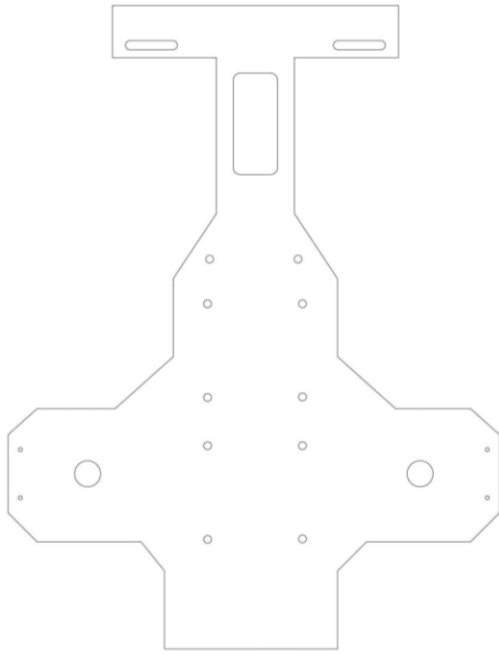
- **Caster Wheel:**

A free-spinning wheel placed at the rear of the chassis to provide balance and support to the robot, especially during turns.



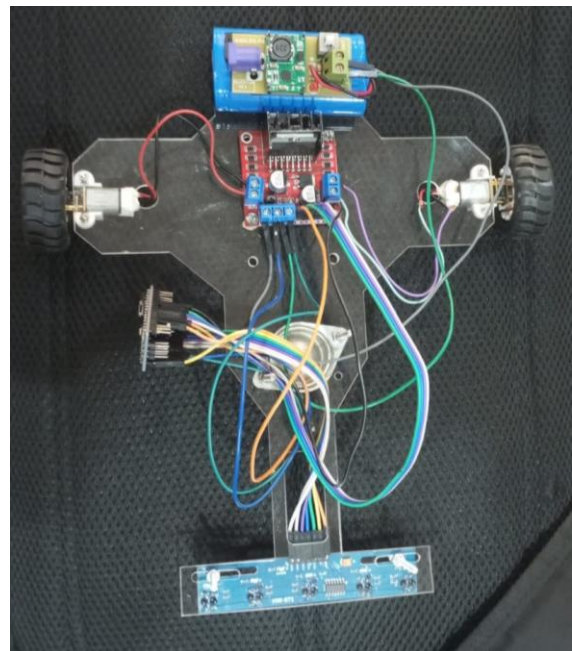
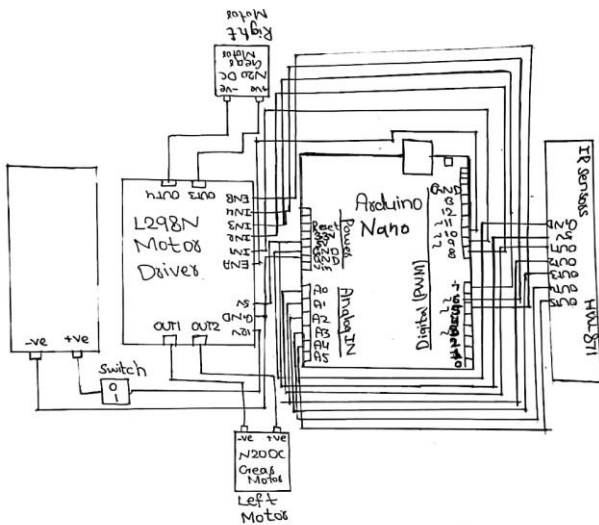
- **Chassis Base:**

A plastic or acrylic platform that physically holds and organizes all components in a compact and stable structure for smooth movement. The chassis was specially designed for this project using solid works which is as follows:



❖ Block Diagram & Real-Life Robot:

Following is the complete Block Diagram For this Project:



❖ Working Principle:

The Line Follower Robot operates on the basic principle of light reflection and detection using infrared (IR) technology. It is designed to follow a black line on a white surface by continuously adjusting its movement based on real-time sensor feedback.

At the heart of the system is a five-channel TCRT5000 IR sensor array, mounted at the front of the robot. Each sensor emits infrared light and detects the amount of reflection from the surface beneath. Since black surfaces absorb infrared light and white surfaces reflect it, the sensors are able to distinguish between the line (black) and the background (white). When a sensor is above a white surface, it outputs a HIGH signal, and when it is above a black line, it outputs a LOW signal.

The sensor outputs are connected to the Arduino Nano, which is programmed to interpret the readings and determine the position of the black line relative to the robot. Based on the combination of HIGH and LOW signals received from the five sensors, the Arduino executes one of several movement commands:

- **Forward Movement:** When the middle sensor (S3) detects the black line, and side sensors read white, the robot moves straight.
- **Slight Turns:** If the black line shifts slightly left or right (detected by S2 or S4), the robot adjusts its direction accordingly by reducing speed on one motor and increasing on the other.
- **Sharp Turns:** If the line is detected on extreme left (S1) or right (S5), the robot performs a sharper turn by reversing one motor and driving the other forward.
- **Stop:** When none of the sensors detect the black line (all sensors HIGH), the robot assumes the line is lost and stops to prevent uncontrolled movement.

The L298N motor driver receives commands from the Arduino and drives the N20 DC gear motors connected to the wheels. The driver enables both direction control (forward/reverse) and speed control using PWM signals from the Arduino.

In summary, the robot continuously:

1. Reads sensor data.
2. Determines the direction of the line.
3. Sends commands to the motors to move, turn, or stop.

This loop allows the robot to autonomously follow a line with high accuracy and responsiveness.

❖ Project Source Code:

Following is the source code for this project:

```
// Motor pins
const int IN1 = 9;
const int IN2 = 8;
const int IN3 = 7;
const int IN4 = 6;
const int ENA = 10; // Left motor speed
const int ENB = 5;  // Right motor speed

// IR sensors (A0-A4)
const int S1 = A0;
const int S2 = A1;
const int S3 = A2;
const int S4 = A3;
const int S5 = A4;

// Set speed
int speedMotor = 100;

void setup() {
  pinMode(IN1, OUTPUT); pinMode(IN2, OUTPUT);
  pinMode(IN3, OUTPUT); pinMode(IN4, OUTPUT);
  pinMode(ENA, OUTPUT); pinMode(ENB, OUTPUT);

  pinMode(S1, INPUT);
  pinMode(S2, INPUT);
  pinMode(S3, INPUT);
  pinMode(S4, INPUT);
  pinMode(S5, INPUT);
}

void loop() {
  // Read sensors
  int s1 = digitalRead(S1);
  int s2 = digitalRead(S2);
  int s3 = digitalRead(S3);
  int s4 = digitalRead(S4);
  int s5 = digitalRead(S5);

  // Logic decision
  if (s2 == 0 && s3 == 1 && s4 == 0) {
    forward();
  }
  else if (s1 == 0 && s2 == 1 && s3 == 1) {
    slightLeft();
  }
  else if (s3 == 1 && s4 == 1 && s5 == 0) {
    slightRight();
  }
  else if (s1 == 1 && s2 == 1) {
    sharpLeft();
  }
  else if (s4 == 1 && s5 == 1) {
    sharpRight();
  }
  else if (s1 == 0 && s2 == 0 && s3 == 0 && s4 == 0 && s5 == 0) {
    stopMotors();
  }
  else {
    forward(); // default action
  }
}

// Motor control functions
void forward() {
  digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW);
  digitalWrite(IN3, HIGH); digitalWrite(IN4, LOW);
  analogWrite(ENA, speedMotor);
}
```

```

    analogWrite(ENB, speedMotor);
}

void slightLeft() {
    digitalWrite(IN1, LOW); digitalWrite(IN2, HIGH); // Left motor slow
    digitalWrite(IN3, HIGH); digitalWrite(IN4, LOW); // Right motor fast
    analogWrite(ENA, speedMotor / 2);
    analogWrite(ENB, speedMotor);
}

void slightRight() {
    digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW); // Left motor fast
    digitalWrite(IN3, LOW); digitalWrite(IN4, HIGH); // Right motor slow
    analogWrite(ENA, speedMotor);
    analogWrite(ENB, speedMotor / 2);
}

void sharpLeft() {
    digitalWrite(IN1, LOW); digitalWrite(IN2, HIGH);
    digitalWrite(IN3, HIGH); digitalWrite(IN4, LOW);
    analogWrite(ENA, speedMotor);
    analogWrite(ENB, speedMotor);
}

void sharpRight() {
    digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW);
    digitalWrite(IN3, LOW); digitalWrite(IN4, HIGH);
    analogWrite(ENA, speedMotor);
    analogWrite(ENB, speedMotor);
}

void stopMotors() {
    digitalWrite(IN1, LOW); digitalWrite(IN2, LOW);
    digitalWrite(IN3, LOW); digitalWrite(IN4, LOW);
    analogWrite(ENA, 0);
    analogWrite(ENB, 0);
}

```

❖ Code Explanation:

The provided Arduino code is programmed to control a line follower robot using a five-channel TCRT5000 IR sensor array and two N20 gear motors driven by an L298N motor driver. In the setup, five IR sensors (S1–S5) are connected to the analog pins (A0–A4) of the Arduino, while the motor driver inputs (IN1–IN4) and speed control pins (ENA and ENB) are connected to the digital PWM pins. Inside the main loop, the robot continuously reads the digital states of the IR sensors to detect the position of the black line. Based on different sensor combinations, it executes one of several predefined motor functions—moving forward, turning slightly or sharply left or right, or stopping entirely if the line is lost. The use of `analogWrite()` allows for speed control, and each direction function manipulates the motor driver's input pins accordingly. This logic enables the robot to follow a path autonomously by responding in real time to changes in the line's position.

❖ **Applications:**

Line follower robots have widespread applications in modern automation and robotics. They are commonly used in industrial automation for transporting goods along predefined paths in warehouses and assembly lines. In education, they serve as an excellent learning platform for introducing students to embedded systems, sensor integration, and motor control. Line followers are also employed in autonomous delivery systems, hospital service robots, and smart traffic management prototypes to demonstrate path-following capabilities.

❖ **Conclusion:**

This project successfully demonstrates the design and implementation of an autonomous line follower robot using fundamental components such as an Arduino microcontroller, IR sensor array, L298N motor driver, and DC gear motors. By integrating sensor-based decision logic with real-time motor control, the robot is able to detect and follow a black path with high responsiveness. Through both hardware and software development, this project not only deepens practical understanding of electronics and embedded programming but also reflects real-world applications of robotics in automation and control systems. The structured use of PWM for speed regulation, logic-based directional control, and effective sensor integration exemplifies a functional and educational robotics solution. Overall, the project serves as a valuable step toward building more intelligent and capable autonomous robotic systems.